

REFERENCES

- [1] Godfray, H. C. J., J. R. Beddington, I. R. Crute, L. Haddad, D. Lawrence, J. F. Muir, J. Pretty, S. Robinson, S. M. Thomas, and C. Toulmin (2010). Food security: The challenge of feeding 9 billion people. *Science*, 327(5967), 812–818.
- [2] Merino, G., M. Barange, J. L. Blanchard, J. Harle, R. Holmes, I. Allen, E. H. Allison, M. C. Badjeck, N. K. Dulvy, J. Holt, S. Jennings, C. Mullon, and L. D. Rodwell (2012). Can marine fisheries and aquaculture meet fish demand from a growing human population in a changing climate? *Global Environmental Change*, 22(4), 795–806.
- [3] Jennings, S., G. D. Stentiford, A. M. Leocadio, K. R. Jeffery, *et al.* (2016). Aquatic food security: insights into challenges and solutions from an analysis of interactions between fisheries, aquaculture, food safety, human health, fish and human welfare, economy and environment. *Fish and Fisheries*, 17(4), 893–938.
- [4] Boyd, C. E. and C. S. Tucker, *Pond Aquaculture Water Quality Management*. Springer US, 1998.
- [5] Ayele, H. S. and M. Atlabachew (2021). Review of characterization, factors, impacts, and solutions of lake eutrophication: lesson for lake Tana, Ethiopia. *Environmental Science and Pollution Research*, 28(12), 14233–14252.
- [6] Martos-Sitcha, J. A., J. M. Mancera, P. Prunet, and L. J. Magnoni (2020). Editorial: Welfare and [tressors in Fish: Challenges Facing Aquaculture. *Frontiers in Physiology*, 11, 162.
- [7] Dong, J., G. Wang, H. Yan, J. Xu, and X. Zhang (2015). A survey of smart water quality monitoring system. *Environmental Science and Pollution Research*, 22.
- [8] Simbeye, D. S., J. Zhao, and S. Yang (2014). Design and deployment of wireless sensor networks for aquaculture monitoring and control based on virtual instruments. *Computers and Electronics in Agriculture*, 102, 31–42.
- [9] Zhu, X., D. Li, D. He, J. Wang, D. Ma, and F. Li (2010). A remote wireless system for water quality online monitoring in intensive fish culture. *Computers and Electronics in Agriculture*, 71, S3 – S9.
- [10] Parra, L., S. Sendra, L. García, and J. Lloret (2018). Design and Deployment of low-cost Sensors for Monitoring the Water Quality and Fish Behavior in Aquaculture Tanks during the Feeding process. *Sensors*, 18(3), 750.
- [11] Parra, L., G. Lloret, J. Lloret, and M. Rodilla (2018). Physical Sensors for Precision Aquaculture: A Review. *IEEE Sensors Journal*, 18(10), 3915–3923.
- [12] Lambrou, T. P., C. C. Anastasiou, C. G. Panayiotou, and M. M. Polycarpou (2014). A low-cost sensor network for real-time monitoring and contamination detection in drinking water distribution systems. *IEEE Sensors Journal*, 14(8), 2765–2772.
- [13] Li, D. and C. Li (2020). Intelligent aquaculture. *Journal of the World Aquaculture Society*, 51(4), 808–814.
- [14] Hu, Z., R. Li, X. Xia, C. Yu, X. Fan, and Y. Zhao (2020). A method overview in smart aquaculture. *Environmental Monitoring and Assessment*, 192(8).

- [15] Ahuja, S. S., Monitoring water quality, pollution assessment, and remediation to assure sustainability. In S. Ahuja (ed.), *Monitoring Water Quality*. Elsevier, Amsterdam, 2013. ISBN 978-0-444-59395-5, 1–18.
- [16] Parmezan, A. R. S., V. M. Souza, and G. E. Batista (2019). Evaluation of statistical and machine learning models for time series prediction: Identifying the state-of-the-art and the best conditions for the use of each model. *Information Sciences*, 484, 302–337.
- [17] Liu, J., C. Yu, Z. Hu, Y. Zhao, *et al.* (2020). Accurate prediction scheme of water quality in smart mariculture with deep Bi-S-SRU learning network. *IEEE Access*, 8, 24784–24798.
- [18] Xie, J. and Q. Wang (2020). Benchmarking machine learning algorithms on blood glucose prediction for Type I Diabetes in comparison with classical time-series models. *IEEE Transactions on Biomedical Engineering*, 67(11), 3101–3124.
- [19] Qin, Y., D. Song, H. Chen, W. Cheng, G. Jiang, and G. Cottrell (2017). A dual-stage attention-based recurrent neural network for time series prediction. *arXiv preprint arXiv:1704.02971*.
- [20] Hu, Z., Y. Zhang, Y. Zhao, M. Xie, *et al.* (2019). A water quality prediction method based on the deep LSTM network considering correlation in smart mariculture. *Sensors*, 19(6), 1420.
- [21] Shi, B., V. Sreeram, D. Zhao, S. Duan, and J. Jiang (2018). A wireless sensor network-based monitoring system for freshwater fishpond aquaculture. *Biosystems Engineering*, 172, 57–66.
- [22] Lin, Y.-B. and H.-C. Tseng (2019). FishTalk: An IoT-Based Mini Aquarium System. *IEEE Access*, 7, 35457–35469.
- [23] Luna, F. D. V. B., E. de la Rosa Aguilar, J. S. Naranjo, and J. G. Jaguey (2017). Robotic System for Automation of Water Quality Monitoring and Feeding in Aquaculture Shadehouse. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 47(7), 1575–1589.
- [24] Encinas, C., E. Ruiz, J. Cortez, and A. Espinoza, Design and implementation of a distributed IoT system for the monitoring of water quality in aquaculture. In *2017 Wireless Telecommunications Symposium (WTS)*. 2017.
- [25] Schmidt, W., D. Raymond, D. Parish, I. G. Ashton, P. I. Miller, C. J. Campos, and J. D. Shutler (2018). Design and operation of a low-cost and compact autonomous buoy system for use in coastal aquaculture and water quality monitoring. *Aquacultural Engineering*, 80, 28–36. ISSN 0144-8609.
- [26] Ferdoush, S. and X. Li (2014). Wireless Sensor Network System Design Using Raspberry Pi and Arduino for Environmental Monitoring Applications. *Procedia Computer Science*, 34, 103–110.
- [27] Liu, Y. T., B. Y. Lin, X. F. Yue, Z. X. Cai, Z. X. Yang, W. H. Liu, S. Y. Huang, J. L. Lu, J. W. Peng, and J. Y. Chen, A solar powered long range real-time water quality monitoring system by LoRaWAN. In *2018 27th Wireless and Optical Communication Conference (WOCC)*. 2018.
- [28] Pule, M., A. Yahya, and J. Chuma (2017). Wireless sensor networks: A survey on monitoring water quality. *Journal of Applied Research and Technology*, 15(6), 562–570.
- [29] Huan, J., H. Li, F. Wu, and W. Cao (2020). Design of water quality monitoring system for aquaculture ponds based on NB-IoT. *Aquacultural Engineering*, 90,

102088.

- [30] Salim, T. I., T. Haiyunnisa, and H. S. Alam, Design and implementation of water quality monitoring for eel fish aquaculture. *In 2016 International Symposium on Electronics and Smart Devices (ISESD)*. IEEE, 2016.
- [31] Luna, M., I. Llorente, and Ángel Cobo (2019). Integration of environmental sustainability and product quality criteria in the decision-making process for feeding strategies in seabream aquaculture companies. *Journal of Cleaner Production*, 217, 691–701.
- [32] Raju, K. R. S. R. and G. H. K. Varma, Knowledge Based Real Time Monitoring System for Aquaculture using IoT. *In 2017 IEEE 7th International Advance Computing Conference (IACC)*. 2017.
- [33] Patrizi, G., A. Bartolini, L. Ciani, V. Gallo, P. Sommella, and M. Carratù (2022). A Virtual Soil Moisture Sensor for Smart Farming Using Deep learning. *IEEE Transactions on Instrumentation and Measurement*, 71, 1–11.
- [34] Yao, L., X. Jiang, G. Huang, J. Qian, B. Shen, L. Xu, and Z. Ge (2020). Virtual sensing f-cao content of cement clinker based on incremental deep dynamic features extracting and transferring model. *IEEE Transactions on Instrumentation and Measurement*, 70, 1–10.
- [35] Wang, J., Z. Deng, Z. Meng, and K. Shen (2021). Magnetoresistive sensor error compensation method using geometry-constraint contour scaling. *IEEE Transactions on Instrumentation and Measurement*, 71, 1–9.
- [36] Ali, S., T. Glass, B. Parr, J. Potgieter, and F. Alam (2020). Low cost sensor with IoT LoRaWAN connectivity and machine learning-based calibration for air pollution monitoring. *IEEE Transactions on Instrumentation and Measurement*, 70, 1–11.
- [37] Alsarraj, A., A. ur Rehman, S. B. Belhaouari, K. M. Saoud, and A. Bermak (2021). Hydrogen sulfide sensor: A concept of physical versus virtual sensing. *IEEE Transactions on Instrumentation and Measurement*, 70, 1–13.
- [38] Grbčić, L., L. Kranjčević, and S. Družeta (2021). Machine learning and simulation-optimization coupling for water distribution network contamination source detection. *Sensors*, 21(4), 1157.
- [39] Béné, C., R. Arthur, H. Norbury, E. H. Allison, M. Beveridge, S. Bush, L. Campling, W. Leschen, D. Little, D. Squires, *et al.* (2016). Contribution of fisheries and aquaculture to food security and poverty reduction: assessing the current evidence. *World Development*, 79, 177–196.
- [40] Liu, S., H. Tai, Q. Ding, D. Li, L. Xu, and Y. Wei (2013). A hybrid approach of support vector regression with genetic algorithm optimization for aquaculture water quality prediction. *Mathematical and Computer Modelling*, 58(3), 458–465.
- [41] hui Yang, Q., B. ping Tan, X. hui Dong, S. yan Chi, and H. yu Liu (2015). Effects of different levels of Yucca schidigera extract on the growth and nonspecific immunity of pacific white shrimp (*Litopenaeus vannamei*) and on culture water quality. *Aquaculture*, 439, 39–44.
- [42] da Silva Burigato Costa, C. M., L. da Silva Marques, A. K. Almeida, I. R. Leite, and I. K. de Almeida (2019). Applicability of water quality models around the world—a review. *Environmental Science and Pollution Research*, 26(36), 36141–36162.
- [43] Arya, F. K. and L. Zhang (2014). Time series analysis of water quality param-

- ters at stillaguamish river using order series method. *Stochastic Environmental Research and Risk Assessment*, 29(1), 227–239.
- [44] Ömer Faruk, D. (2010). A hybrid neural network and ARIMA model for water quality time series prediction. *Engineering Applications of Artificial Intelligence*, 23(4), 586–594.
 - [45] Arya, F. K. and L. Zhang (2015). Time series analysis of water quality parameters at stillaguamish river using order series method. *Stochastic environmental research and risk assessment*, 29(1), 227–239.
 - [46] Kashyap, R. L. (1982). Optimal choice of AR and MA parts in autoregressive moving average models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-4(2), 99–104.
 - [47] Saboia, J. L. M. (1977). Autoregressive integrated moving average (ARIMA) models for birth forecasting. *Journal of the American Statistical Association*, 72(358), 264–270.
 - [48] Jeong, K., C. Koo, and T. Hong (2014). An estimation model for determining the annual energy cost budget in educational facilities using SARIMA (seasonal autoregressive integrated moving average) and ANN (artificial neural network). *Energy*, 71, 71–79.
 - [49] Zhang, H., X. Wang, J. Cao, M. Tang, and Y. Guo (2018). A multivariate short-term traffic flow forecasting method based on wavelet analysis and seasonal time series. *Applied Intelligence*, 48(10), 3827–3838.
 - [50] Sasikala, D. and K. V. Sharma (2022). Water quality prediction and classification based on linear discriminant analysis and light gradient boosting machine classifier approach. *Journal of Innovative Image Processing*, 4(3), 147–156.
 - [51] Jere, S., A. Banda, B. Kasense, I. Siluyele, and E. Moyo (2019). Forecasting annual international tourist arrivals in Zambia using Holt-Winters exponential smoothing. *Open Journal of Statistics*, 9(2), 258–267.
 - [52] Litterman, R. B. (1983). A random walk, Markov model for the distribution of time series. *Journal of Business & Economic Statistics*, 1(2), 169–173.
 - [53] Tseng, F.-M., H.-C. Yu, and G.-H. Tzeng (2001). Applied hybrid grey model to forecast seasonal time series. *Technological Forecasting and Social Change*, 67(2-3), 291–302.
 - [54] Pai, P. F., K. P. Lin, C. S. Lin, and P. T. Chang (2010). Time series forecasting by a seasonal support vector regression model. *Expert Systems with Applications*, 37(6), 4261–4265.
 - [55] Liu, M. D., L. Ding, and Y. L. Bai (2021). Application of hybrid model based on empirical mode decomposition, novel recurrent neural networks and the ARIMA to wind speed prediction. *Energy Conversion and Management*, 233, 113917.
 - [56] Rehana, S., River water temperature modelling under climate change using support vector regression. In *Hydrology in a changing world*. Springer, 2019, 171–183.
 - [57] Liu, S., H. Tai, Q. Ding, D. Li, L. Xu, and Y. Wei (2013). A hybrid approach of support vector regression with genetic algorithm optimization for aquaculture water quality prediction. *Mathematical and Computer Modelling*, 58(3-4), 458–465.
 - [58] Barzegar, R., J. Adamowski, and A. A. Moghaddam (2016). Application of wavelet-artificial intelligence hybrid models for water quality prediction: a

- case study in Aji-Chay River, Iran. *Stochastic environmental research and risk assessment*, 30(7), 1797–1819.
- [59] Heddam, S. and O. Kisi (2017). Extreme learning machines: a new approach for modeling dissolved oxygen (DO) concentration with and without water quality variables as predictors. *Environmental Science and Pollution Research*, 24(20), 16702–16724.
 - [60] Li, X., H. Xie, R. Wang, Y. Cai, J. Cao, F. Wang, H. Min, and X. Deng (2016). Empirical analysis: Stock market prediction via extreme learning machine. *Neural Computing and Applications*, 27(1), 67–78.
 - [61] Zhang, X., S. Qu, J. Huang, B. Fang, and P. Yu (2018). Stock market prediction via multi-source multiple instance learning. *IEEE Access*, 6, 50720–50728.
 - [62] Chen, G., C. Ding, Y. Li, X. Hu, X. Li, L. Ren, X. Ding, P. Tian, and W. Xue (2020). Prediction of Chronic Kidney Disease using Adaptive Hybridized Deep Convolutional Neural Network on the Internet of Medical Things Platform. *IEEE Access*, 8(99), 100497–100508.
 - [63] Contreras, J., R. Espinola, F. Nogales, and A. Conejo (2003). ARIMA models to predict next-day electricity prices. *IEEE Transactions on Power Systems*, 18(3), 1014–1020.
 - [64] Arcomano, T., I. Szunyogh, J. Pathak, A. Wikner, B. R. Hunt, and E. Ott (). A machine learning-based global atmospheric forecast model. *Geophysical Research Letters*, 47(9), e2020GL087776.
 - [65] Yu, Y., J. Cao, and J. Zhu (2019). An LSTM short-term solar irradiance forecasting under complicated weather conditions. *IEEE Access*, 7, 145651–145666.
 - [66] Singh, S. (2008). A computational method of forecasting based on fuzzy time series. *Mathematics and Computers in Simulation*, 79(3), 539–554.
 - [67] Li, L., P. Jiang, H. Xu, G. Lin, D. Guo, and H. Wu (2019). Water quality prediction based on recurrent neural network and improved evidence theory: a case study of Qiantang River, China. *Environmental Science and Pollution Research*, 26(19), 19879–19896.
 - [68] Xu, L. and S. Liu (2013). Study of short-term water quality prediction model based on wavelet neural network. *Mathematical and Computer Modelling*, 58(3-4), 807–813.
 - [69] Sun, X., W. Sun, J. Wang, Y. Zhang, and Y. Gao (2016). Using a grey-markov model optimized by Cuckoo search algorithm to forecast the annual foreign tourist arrivals to China. *Tourism Management*, 52, 369–379.
 - [70] Naug, A. and G. Biswas, A data driven method for prediction of energy demand in commercial buildings. In *2018 IEEE 14th International Conference on Automation Science and Engineering (CASE)*. IEEE, 2018.
 - [71] Avila, R., B. Horn, E. Moriarty, R. Hodson, and E. Moltchanova (2018). Evaluating statistical model performance in water quality prediction. *Journal of Environmental Management*, 206, 910–919.
 - [72] Zou, S. and Y.-S. Yu (1996). A dynamic factor model for multivariate water quality time series with trends. *Journal of Hydrology*, 178(1-4), 381–400.
 - [73] Wu, J., J. Lu, and J. Wang (2009). Application of chaos and fractal models to water quality time series prediction. *Environmental Modelling & Software*, 24(5), 632–636.

- [74] Feng, T., C. Wang, J. Hou, P. Wang, Y. Liu, Q. Dai, Y. Yang, and G. You (2018). Effect of inter-basin water transfer on water quality in an urban lake: A combined water quality index algorithm and biophysical modelling approach. *Ecological Indicators*, 92, 61–71.
- [75] Chan, S., W. Thoe, and J. Lee (2013). Real-time forecasting of hong kong beach water quality by 3d deterministic model. *Water Research*, 47(4), 1631–1647.
- [76] Cho, S., B. Lim, J. Jung, S. Kim, H. Chae, J. Park, S. Park, and J. K. Park (2014). Factors affecting algal blooms in a man-made lake and prediction using an artificial neural network. *Measurement*, 53, 224–233.
- [77] Ümmühan Başaran Filik and T. Filik (2017). Wind speed prediction using artificial neural networks based on multiple local measurements in eskisehir. *Energy Procedia*, 107, 264–269.
- [78] Tan, G., J. Yan, C. Gao, and S. Yang (2012). Prediction of water quality time series data based on least squares support vector machine. *Procedia Engineering*, 31, 1194–1199.
- [79] Singh, K. P., N. Basant, and S. Gupta (2011). Support vector machines in water quality management. *Analytica Chimica Acta*, 703(2), 152–162.
- [80] Preis, A. and A. Ostfeld (2008). A coupled model tree–genetic algorithm scheme for flow and water quality predictions in watersheds. *Journal of Hydrology*, 349(3-4), 364–375.
- [81] Najafabadi, M. M., F. Villanustre, T. M. Khoshgoftaar, N. Seliya, R. Wald, and E. Muharemagic (2015). Deep Learning applications and challenges in big data analytics. *Journal of Big Data*, 2(1), 1–21.
- [82] Xu, X. and M. Yoneda (2021). Multitask air-quality prediction based on LSTM-autoencoder model. *IEEE Transactions on Cybernetics*, 51(5), 2577–2586.
- [83] Solgi, R., H. A. Loaiciga, and M. Kram (2021). Long short-term memory neural network (LSTM-nn) for aquifer level time series forecasting using in-situ piezometric observations. *Journal of Hydrology*, 601, 126800.
- [84] Chandra, B. and R. K. Sharma, On improving recurrent neural network for image classification. In *2017 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2017.
- [85] Liu, P., J. Wang, A. Sangaiah, Y. Xie, and X. Yin (2019). Analysis and Prediction of Water Quality Using LSTM Deep Neural Networks in IoT Environment. *Sustainability*, 11(7), 2058.
- [86] Hu, Z., Y. Zhang, Y. Zhao, M. Xie, J. Zhong, Z. Tu, and J. Liu (2019). A water quality prediction method based on the deep LSTM network considering correlation in smart mariculture. *Sensors*, 19(6), 1420.
- [87] Gers, F. A., J. Schmidhuber, and F. Cummins (2000). Learning to forget: Continual prediction with LSTM. *Neural Computation*, 12(10), 2451–2471.
- [88] Chang, Z., Y. Zhang, and W. Chen, Effective adam-optimized LSTM neural network for electricity price forecasting. In *2018 IEEE 9th international conference on software engineering and service science (ICSESS)*. IEEE, 2018.
- [89] Zhang, T., S. Song, S. Li, L. Ma, *et al.* (2019). Research on Gas concentration prediction models based on LSTM multidimensional time series. *Energies*, 12(1), 161.
- [90] Duchi, J., E. Hazan, and Y. Singer (2011). Adaptive subgradient methods for online learning and stochastic optimization. *Journal of Machine Learning*

- Research*, 12(7).
- [91] De, S., A. Mukherjee, and E. Ullah (2018). Convergence guarantees for rmsprop and adam in non-convex optimization and an empirical comparison to nesterov acceleration. *arXiv preprint arXiv:1807.06766*.
 - [92] Huang, C.-J., Y. Shen, Y.-H. Chen, and H.-C. Chen (2021). A novel hybrid deep neural network model for short-term electricity price forecasting. *International Journal of Energy Research*, 45(2), 2511–2532.
 - [93] Sidhom, J.-W., H. B. Larman, D. M. Pardoll, and A. S. Baras (2021). Deeptcr is a Deep Learning framework for revealing sequence concepts within t-cell repertoires. *Nature communications*, 12(1), 1–12.
 - [94] LeCun, Y., Y. Bengio, and G. Hinton (2015). Deep Learning. *Nature*, 521(7553), 436–444.
 - [95] Yuan, Y., M. Chao, and Y.-C. Lo (2017). Automatic skin lesion segmentation using deep fully convolutional networks with jaccard distance. *IEEE Transactions on Medical Imaging*, 36(9), 1876–1886.
 - [96] Kourkounakis, T., A. Hajavi, and A. Etemad (2021). Fluentnet: End-to-end detection of stuttered speech disfluencies with Deep Learning. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 29, 2986–2999.
 - [97] Zhao, J., F. Huang, J. Lv, Y. Duan, Z. Qin, G. Li, and G. Tian, Do RNN and LSTM have long memory? In H. D. III and A. Singh (eds.), *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*. PMLR, 2020.
 - [98] Zheng, W. and G. Chen (2021). An Accurate GRU-based Power Time-Series Prediction approach with Selective State Updating and Stochastic Optimization. *IEEE Transactions on Cybernetics*, 1–13.
 - [99] Zheng, W., P. Zhao, K. Huang, and G. Chen, Understanding the property of long term memory for the LSTM with attention mechanism. In *Proceedings of the 30th ACM International Conference on Information & Knowledge Management*. ACM, 2021.
 - [100] Guo, T., T. Lin, and N. Antulov-Fantulin, Exploring interpretable LSTM neural networks over multi-variable data. In *International conference on machine learning*. PMLR, 2019.
 - [101] Song, D., H. Chen, G. Jiang, and Y. Qin (2021). Dual stage attention based recurrent neural network for time series prediction.
 - [102] Sun, Q., M. V. Jankovic, L. Bally, and S. G. Mougiakakou, Predicting blood glucose with an LSTM and bi-LSTM based deep neural network. In *2018 14th symposium on neural networks and applications (NEUREL)*. IEEE, 2018.
 - [103] Zhang, Y., B. Yan, and M. Aasma (2020). A novel Deep Learning framework: Prediction and analysis of financial time series using ceemd and LSTM. *Expert Systems with Applications*, 159, 113609.
 - [104] Sunny, M. A. I., M. M. S. Maswood, and A. G. Alharbi, Deep Learning-based stock price prediction using LSTM and bi-directional LSTM model. In *2020 2nd Novel Intelligent and Leading Emerging Sciences Conference (NILES)*. IEEE, 2020.
 - [105] Gul, M. J., G. M. Urfa, A. Paul, J. Moon, S. Rho, and E. Hwang (2021). Mid-term electricity load prediction using CNN and Bi-LSTM. *The Journal of Supercomputing*, 77(10), 10942–10958.
 - [106] Hewage, P., M. Trovati, E. Pereira, and A. Behera (2021). Deep Learning-

- based effective fine-grained weather forecasting model. *Pattern Analysis and Applications*, 24(1), 343–366.
- [107] Peng, T., C. Zhang, J. Zhou, and M. S. Nazir (2021). An integrated framework of bi-directional long-short term memory (BiLSTM) based on sine cosine algorithm for hourly solar radiation forecasting. *Energy*, 221, 119887.
 - [108] Ma, J., J. C. Cheng, C. Lin, Y. Tan, and J. Zhang (2019). Improving air quality prediction accuracy at larger temporal resolutions using Deep Learning and transfer learning techniques. *Atmospheric Environment*, 214, 116885.
 - [109] Singhal, R., S. Srivatsan, and P. Panda (2022). A novel multimodal method for depression identification. *Journal of Trends in Computer Science and Smart Technology*, 4(4), 215–225.
 - [110] Du, Y., J. Li, C. Li, and P. Duan (2022). A reinforcement learning approach for flexible job shop scheduling problem with crane transportation and setup times. *IEEE Transactions on Neural Networks and Learning Systems*, 1–15.
 - [111] Du, Y., J.-q. Li, X.-l. Chen, P.-y. Duan, and Q.-k. Pan (2022). Knowledge-Based Reinforcement Learning and Estimation of Distribution Algorithm for Flexible Job Shop Scheduling Problem. *IEEE Transactions on Emerging Topics in Computational Intelligence*, 1–15.
 - [112] Qin, Y., D. Song, H. Chen, W. Cheng, G. Jiang, and G. Cottrell (2017). A dual-stage attention-based recurrent neural network for time series prediction. *arXiv preprint arXiv:1704.02971*.
 - [113] Liang, Y., S. Ke, J. Zhang, X. Yi, and Y. Zheng, GeoMAN: Multi-level attention networks for geo-sensory time series prediction. *In Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence, IJCAI-18*. International Joint Conferences on Artificial Intelligence Organization, 2018.
 - [114] Gholamalinezhad, H. and H. Khosravi (2020). Pooling methods in deep neural networks, a review. *arXiv preprint arXiv:2009.07485*.
 - [115] Bradbury, J., S. Merity, C. Xiong, and R. Socher (2016). Quasi-recurrent neural networks. *arXiv preprint arXiv:1611.01576*.
 - [116] Lien Minh, D., A. Sadeghi-Niaraki, H. D. Huy, K. Min, and H. Moon (2018). Deep Learning approach for short-term stock trends prediction based on two-stream gated recurrent unit network. *IEEE Access*, 6, 55392–55404.
 - [117] Tanwar, S., N. P. Patel, S. N. Patel, J. R. Patel, G. Sharma, and I. E. Davidson (2021). Deep Learning-based cryptocurrency price prediction scheme with inter-dependent relations. *IEEE Access*, 9, 138633–138646.
 - [118] Livieris, I. E., E. Pintelas, and P. Pintelas (2020). A CNN–LSTM model for gold price time-series forecasting. *Neural Computing and Applications*, 32(23), 17351–17360.
 - [119] Li, K., J. Daniels, C. Liu, P. Herrero, and P. Georgiou (2020). Convolutional recurrent neural networks for glucose prediction. *IEEE Journal of Biomedical and Health Informatics*, 24(2), 603–613.
 - [120] Du, S., T. Li, Y. Yang, and S.-J. Horng (2021). Deep air quality forecasting using hybrid Deep Learning framework. *IEEE Transactions on Knowledge and Data Engineering*, 33(6), 2412–2424.
 - [121] Sharma, E., R. C. Deo, R. Prasad, A. V. Parisi, and N. Raj (2020). Deep air quality forecasts: Suspended particulate matter modeling with convolutional neural and long short-term memory networks. *IEEE Access*, 8, 209503–

209516.

- [122] Chiang, P.-W. and S.-J. Horng (2021). Hybrid time-series framework for daily-based PM_{2.5} forecasting. *IEEE Access*, 9, 104162–104176.
- [123] Liu, Y., C. Gong, L. Yang, and Y. Chen (2020). DSTP-RNN: A dual-stage two-phase attention-based recurrent neural network for long-term and multivariate time series prediction. *Expert Systems with Applications*, 143, 113082.
- [124] Hou, C., J. Wu, B. Cao, and J. Fan (2021). A deep-learning prediction model for imbalanced time series data forecasting. *Big Data Mining and Analytics*, 4(4), 266–278.
- [125] Dairi, A., F. Harrou, S. Khadraoui, and Y. Sun (2021). Integrated multiple directed attention-based Deep Learning for improved air pollution forecasting. *IEEE Transactions on Instrumentation and Measurement*, 70, 1–15.
- [126] Jung, S., J. Moon, S. Park, and E. Hwang (2021). Self-attention-based Deep Learning network for regional influenza forecasting. *IEEE Journal of Biomedical and Health Informatics*, 26(2), 922–933.
- [127] Ardjosoediro, I. and I. W. Ramnarine (2002). The influence of turbidity on growth, feed conversion and survivorship of the Jamaica red tilapia strain. *Aquaculture*, 212(1-4), 159–165.
- [128] Liu, S., L. Xu, D. Li, Q. Li, Y. Jiang, H. Tai, and L. Zeng (2013). Prediction of dissolved oxygen content in river crab culture based on least squares support vector regression optimized by improved particle swarm optimization. *Computers and Electronics in Agriculture*, 95, 82–91.
- [129] Rubio, V., F. Sánchez-Vázquez, and J. Madrid (2005). Effects of salinity on food intake and macronutrient selection in european sea bass. *Physiology & Behavior*, 85(3), 333–339.
- [130] Buentello, J., D. M. Gatlin, and W. H. Neill (2000). Effects of water temperature and dissolved oxygen on daily feed consumption, feed utilization and growth of channel catfish (*ictalurus punctatus*). *Aquaculture*, 182(3), 339–352.
- [131] Svobodová, Z., R. Lloyd, and J. Machova, *Causes and effects of pollution of fish*. Food and Agriculture Organization of the United Nations, 1993, 7–11.
- [132] dos Santos Simões, F., A. B. Moreira, M. C. Bisinoti, S. M. N. Gimenez, and M. J. S. Yabe (2008). Water quality index as a simple indicator of aquaculture effects on aquatic bodies. *Ecological Indicators*, 8(5), 476–484.
- [133] Zdenka Svobodová, J. M. B. V., Richard Lloyd, *Water quality and fish health*. Food and Agriculture Organization of the United Nations Rome, 1993.
- [134] Weiss, R., The solubility of nitrogen, oxygen and argon in water and seawater. *In 1970, Deep-Sea Research*, volume 17. 1970.
- [135] Benson, B. B. and J. Daniel Krause, The concentration and isotopic fractionation of gases dissolved in freshwater in equilibrium with the atmosphere. 1. oxygen. *In 1980, by The American Society of Limnology and Oceanography, Inc.*. 1980.
- [136] Benson, B. B. and J. Daniel Krause, The concentration and isotopic fractionation of oxygen dissolved in freshwater and seawater in equilibrium with the atmosphere. *In 1984, by The American Society of Limnology and Oceanography, Inc.*. 1984.
- [137] Garcia, H. E. and L. I. Gordon, Oxygen Solubility in Seawater: Better Fitting Equations. *In 1992, by the American Society of Limnology and Oceanography, Inc.*, volume 37. 1992.

- [138] Hochreiter, S. and J. Schmidhuber (1997). Long Short-Term Memory. *Neural Computation*, 9(8), 1735–1780.
- [139] Cho, K., B. Van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio (2014). Learning phrase representations using rnn encoder-decoder for statistical machine translation. *arXiv preprint arXiv:1406.1078*.
- [140] PAN, L.-Q., J.-Z. LI, and J.-Z. LUO (2010). A temporal and spatial correlation based missing values imputation algorithm in wireless sensor networks. *Chinese Journal of Computers*, 33(1), 1–11.
- [141] Liu, P., J. Wang, A. Sangaiah, Y. Xie, *et al.* (2019). Analysis and prediction of water quality using LSTM deep neural networks in IoT environment. *Sustainability*, 11(7), 2058.
- [142] Yu, Y., X. Si, C. Hu, and J. Zhang (2019). A review of recurrent neural networks: LSTM cells and network architectures. *Neural Computation*, 31(7), 1235–1270.
- [143] Wu, K., J. Wu, L. Feng, B. Yang, R. Liang, S. Yang, and R. Zhao (2021). An attention-based CNN-LSTM-BiLSTM model for short-term electric load forecasting in integrated energy system. *International Transactions on Electrical Energy Systems*, 31(1), e12637.
- [144] Siami Namini, S., N. Tavakoli, and A. Siami Namin, The performance of LSTM and BiLSTM in Forecasting Time Series. 2019.
- [145] Qiu, H., C. Fan, J. Yao, and X. Ye (2020). Chinese Microblog Sentiment Detection Based on CNN-BiGRU and Multihead Attention Mechanism. *Scientific Programming*, 2020(13).
- [146] Pedregosa, F., G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, *et al.* (2011). Scikit-learn: Machine learning in Python. *The Journal of Machine Learning research*, 12, 2825–2830.
- [147] Kingma, D. P. and J. Ba (2017). Adam: A method for Stochastic Optimization.
- [148] Barzegar, R., M. T. Aalami, and J. Adamowski (2020). Short-term water quality variable prediction using a hybrid CNN-LSTM Deep Learning model. *Stochastic Environmental Research and Risk Assessment*, 1–19.
- [149] Bahdanau, D., K. Cho, and Y. Bengio (2014). Neural machine translation by jointly learning to align and translate. *arXiv preprint arXiv:1409.0473*.
- [150] Okewu, E., P. Adewole, and O. Sennaiké, Experimental comparison of stochastic optimizers in Deep Learning. *In International Conference on Computational Science and Its Applications*. Springer, 2019.